

SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

Executive Summary

This report compares energy consumption and operational costs between baseline and all applied renovation scenarios from CSV results. Embodied carbon is annualized using an analysis period of 30 years, while operational carbon is reported over 1 year. The construction cost data is from 2026 RSMeans Database. The embodied carbon is calculated using data from EC3 Environmental Product Declaration (EPD) Database. Since RSMeans Database is proprietary, users can adopt customized cost dataset when needed.

Building Information

- Weather File: USA_MN_Duluth.Intl.AP-Duluth.ANGB.727450_TMYx.epw
- Building Name: SmallOffice
- Building Location: Duluth, MN
- Total Floor Area: 511 m²

Renovation Details by Scenario

Scenario	Renovation Type	Renovation Details
Baseline	baseline	Baseline (no envelope renovation)
Scenario 1	wall + roof	Wall upgrade: • Exterior wall insulation: Fiberglass Batts, target R-10.0 (skipped: target R-value already satisfied) Roof upgrade: • Roof insulation: Blown Fiberglass, target R-20.0

Scenario 2	wall	Wall upgrade: Exterior wall insulation: Expanded Polystyrene (EPS) Foam Board, target R-30.0
Scenario 3	wall + roof + window + door	Wall upgrade: - Exterior wall insulation: Fiberglass Batts, target R-20.4 Roof upgrade: - Roof insulation: Blown Fiberglass, target R-34.5 Window upgrade: 3-pane replacement (skipped: SimpleGlazing windows) - Weatherstrip application (skipped: no OperableWindow) - Glazing film application (skipped: SimpleGlazing windows) - Caulking application: acrylic Door upgrade: - Door replacement: wooden door (1 of 3 door(s) skipped: door type incompatibility) - Bottom seal: automatic door bottom - Top/side seal: jamb weatherstrip

Annual Energy Analysis

Scenario	Metric	Baseline	Renovation	Delta	Savings %
Scenario 1	Total Site Energy (GJ)	236.54	234.60	1.94	0.82%
Scenario 2	Total Site Energy (GJ)	236.54	233.46	3.08	1.30%
Scenario 3	Total Site Energy (GJ)	236.54	223.47	13.07	5.53%

Key Performance Metrics

Operational Cost Saving (Scenario - Baseline) (\$/yr) Comparison

Baseline		\$9,841
Scenario 3		\$9,411

■ Baseline ■ Scenario 3

Max saving: **\$-430 (-4.37%)**

Operational Carbon Saving (Scenario - Baseline) (kg CO₂e/yr) Comparison

Baseline		20,894
Scenario 3		20,026

■ Baseline ■ Scenario 3

Max saving: **-868 kg CO₂e (-4.15%)**

Min Retrofit Construction Cost

\$5,241

Scenario 1

Min Retrofit Embodied Carbon

1,577 kg CO₂e

Embodied carbon intensity (ECI): 3.08 kgCO₂e/m²
Scenario 2

Lowest Retrofit Cost Payback Period

36 years

Scenario 3

Lowest Retrofit Carbon Payback Period

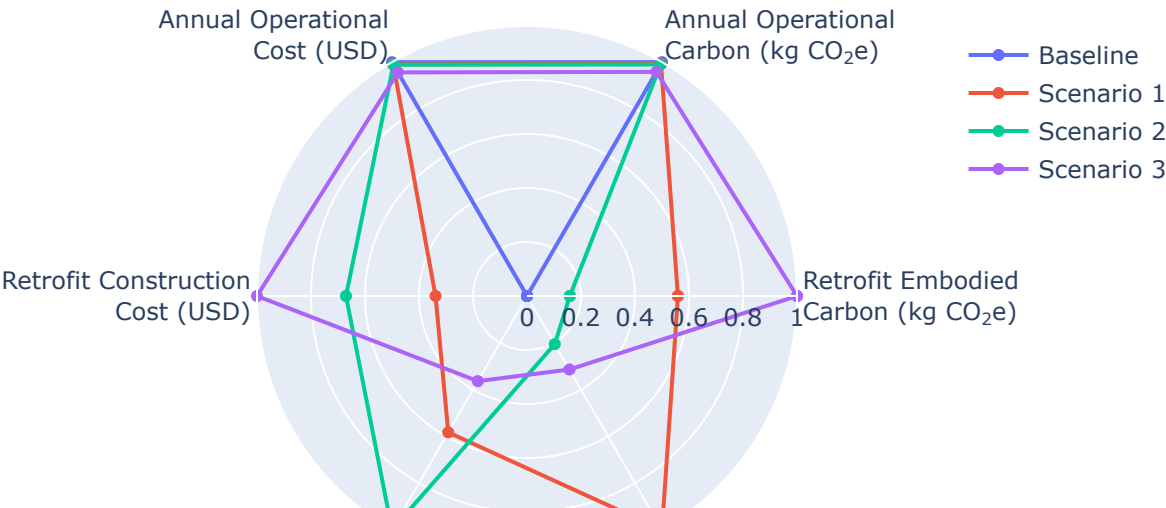
7.49 years

Scenario 2

Comparative Visualizations between Renovation Scenarios

Spider Chart Visualization

Parametric Scenario Comparison (Normalized Spider Chart)





Payback Period of the Retrofit

Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) (\$/yr)).

Scenario 1		58 yrs
Scenario 2		99 yrs
Scenario 3		36 yrs

Carbon Payback Period

Payback = retrofit embodied carbon / abs(Operational Carbon Saving (Scenario - Baseline) (kg CO₂e/yr)).

Scenario 1		36 yrs
Scenario 2		7.49 yrs
Scenario 3		11 yrs

Material List

Material properties and consumption by renovation scenario. Missing values are shown as N/A.

Scenario	Component	Material name	Volume (m ³)	Area (m ²)	Thickness (m)	Length (m)	Thermal Conductivity (W/m·K)	Density (kg/m ³)	Product Lifetime (years)
Scenario 1	Wall insulation	Fiberglass Batts	N/A	N/A	N/A	N/A	0.03	30	60
Scenario 1	Roof insulation	Blown Fiberglass	82	599	0.14	N/A	0.03	30	68
Scenario 2	Wall insulation	Expanded Polystyrene (EPS) Foam Board	21	282	N/A	N/A	0.03	20	60
Scenario 3	Wall insulation	Fiberglass Batts	4.86	282	N/A	N/A	0.03	30	60
Scenario 3	Roof insulation	Blown Fiberglass	144	599	0.24	N/A	0.03	30	68

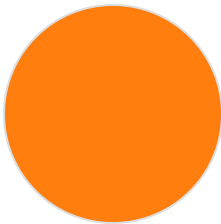
Scenario 3	Window frame	N/A	N/A	56	N/A	N/A	N/A	N/A	15
Scenario 3	Window caulking	acrylic	0.0034	N/A	N/A	N/A	N/A	N/A	10
Scenario 3	Door	wooden door	N/A	3.90	N/A	N/A	0.15	638	30
Scenario 3	Door bottom sealing	automatic door bottom	N/A	N/A	N/A	3.66	N/A	N/A	15
Scenario 3	Door top/side sealing	jamb weatherst rip	N/A	N/A	N/A	16	N/A	N/A	15

Result Summary

Per-scenario retrofit contribution breakdown. Each scenario includes two pie charts: cost and embodied carbon by retrofit measure.

Scenario 1

Cost breakdown by retrofit measure



Roof: \$5,241 (100.0%)

Total: \$5,241

Carbon breakdown by retrofit measure

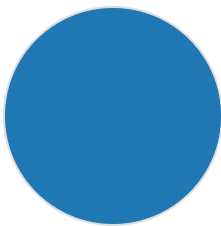


Roof: 5,551 (100.0%)

Total: 5,551 kg CO₂e

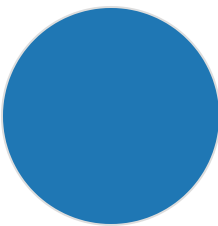
Scenario 2

Cost breakdown by retrofit measure



Wall: \$10,372 (100.0%)

Carbon breakdown by retrofit measure



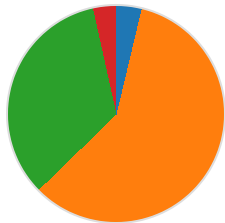
Wall: 1,577 (100.0%)

Total: \$10,372

Total: 1,577 kg CO₂e

Scenario 3

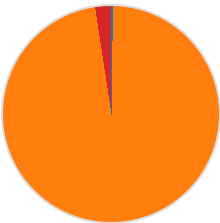
Cost breakdown by retrofit measure



- Wall: \$576 (3.7%)
- Roof: \$9,128 (59.0%)
- Window: \$5,229 (33.8%)
- Door: \$537 (3.5%)

Total: \$15,470

Carbon breakdown by retrofit measure



- Wall: 21 (0.2%)
- Roof: 9,669 (97.4%)
- Door: 241 (2.4%)

Total: 9,932 kg CO₂e

Scenario	Retrofit Embodied Carbon (kg CO ₂ e)	Retrofit Construction Cost (USD)	Annual Operational Carbon (kg CO ₂ e)	Annual Operational Cost (USD)
Baseline	0.00	\$0.00	20,894	\$9,841
Scenario 1	5,551	\$5,241	20,742	\$9,750
Scenario 2	1,577	\$10,372	20,684	\$9,736
Scenario 3	9,932	\$15,470	20,026	\$9,411